

Sai Tarra

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EDUCATION

University of Virginia

Master of Science in Computer Science

Charlottesville, VA

August 2026 – Expected June 2028

University of California, Santa Cruz

Bachelor of Science in Computer Science GPA: 3.82/4.00

Santa Cruz, CA

September 2022 – December 2025

EXPERIENCE

Software Engineer, AI Training

March 2024 – Present

DataAnnotation

- Reviewed and optimized LLM generated code (C, C++, Python and JavaScript) across various use cases to improve accuracy and performance for future model training.
- Created highly specific prompts to solve software engineering problems to stress test models and identify flaws.

PROJECTS

Distributed Key Value Store | *Python, FastAPI, Docker*

- Developed a distributed, causally consistent, fault tolerant, key value store exposed with a REST interface.
- Scales horizontally with improved throughput by placing several nodes into individual shards. Used consistent hashing to easily add/remove nodes from the network.
- High availability by allowing operations to work as long as one replica is reachable through causal consistency.
- Used Docker containers, networks and iptables to test for correctness and performance.

Custom Compiler | *Python, C++, Parsing, Semantic and Syntactic analysis, Code generation*

- Compiles a language (Similar to C) into an LLVM like intermediate representation (Similar to assembly code).
- Wrote a recursive descent parser defined by a context free grammar. Created a type checker and code generator.
- Added optimizations to output IR code such as loop unrolling and local value numbering to improve runtime.

Regex Engine | *C, Parsing, Automata*

- Created a regular expression engine in C supporting all the traditional operators of regex including Concatenation, Union, Kleene star, Groups, Ranges etc.
- Provides a simple interface supporting various methods such as full string matching and start of string matching and aims to be easy to use.

Chaptre | *TypeScript, React, Express, Firebase, Jest*

- Created a story sharing platform where authors can sign up to create and customize a profile in which they can write stories and publish them for people to read.
- Made an editor, rating/feedback system, and social network feature.
- Used SCRUM and Jira to manage a team to get the project done on time.

Multithreaded HTTP Server | *C, Sockets, Multithreading*

- Developed a multithreaded web server capable of serving static websites with support for HTTP 1.1 requests with methods like PUT and GET.
- Used Unix sockets to handle networking and pthreads to create a customizable thread pool to improve throughput of requests.

Custom Emulator | *C++, SDL2, CMake*

- Built an emulator in C++ for a custom game console with its own CPU and GPU instruction set.
- Created a pipeline to execute the binary ROM files. Added input with memory mapped IO and added visual output with SDL2.

TECHNICAL SKILLS

Languages: C, C++, Python, Javascript, SQL, HTML/CSS

Tools/Frameworks/Libraries: Linux, Git, Docker, Pandas, Express.js, Flask, React, Firebase, FastAPI

Courses/Skills: Compilers, Operating Systems, Distributed Systems, Multithreading, Data structures & Algorithms, Object Oriented Programming, Computer Architecture, Databases